

LOW ACRYLONITRILE NBR PATENT LANDSCAPE: COMPOSITION CONTROL, CONTINUOUS FEED STRATEGIES, AND EMULSION POLYMERIZATION ROUTES

1. ABSTRACT

This report synthesizes patent evidence relevant to low-acrylonitrile nitrile butadiene rubber (NBR), with emphasis on how acrylonitrile content is controlled during copolymer synthesis and how process design affects composition uniformity, conversion, and downstream properties. The target material is low-ACN NBR, generally understood as NBR with reduced acrylonitrile content relative to standard grades, often in the approximate 10–18 wt% range and typically below about 20 wt% ACN. The patent set spans three distinct but related approaches: a modern controlled emulsion/block-copolymer platform that explicitly incorporates nitrile-rubber blocks containing acrylonitrile and butadiene; a continuous emulsion copolymerization process designed to manage monomer ratio drift and achieve high bound acrylonitrile; and an older grafting/latex-based route that uses preformed NBR latex as a rubber phase in impact-resistant nitrile polymer compositions.

Across the landscape, the dominant technical theme is monomer composition control. US5770660A shows that incremental butadiene addition in multiple reactors can maintain a more constant acrylonitrile/butadiene ratio and produce NBR with about 48.8–59.5 wt% bound acrylonitrile, demonstrating the importance of feed staging and conversion management. US7592409B2 is not a direct low-ACN NBR patent, but it is relevant because it discloses emulsion polymerization with in-situ emulsifier formation and block architectures that include a nitrile-rubber B block composed of acrylonitrile and 1,3-butadiene; this supports controlled emulsion methods that could be adapted to low-ACN NBR block or seed-latex synthesis. US3426102A discloses preformed NBR latexes with acrylonitrile/butadiene ratios such as 40/60 and 25/75 used as impact modifiers in aqueous polymerization, illustrating early latex-based NBR preparation and the role of low-to-moderate ACN rubber phases in property tuning.

Overall, the evidence favors emulsion-based synthesis, careful monomer feed scheduling, and use of emulsifier/initiator systems that preserve latex stability while controlling composition drift. For low-ACN NBR, the most transferable parameters are staged monomer addition, moderate temperatures in the 45–60 C range for conventional emulsion routes, and maintaining a stable aqueous phase with suitable emulsifier and chain-transfer control.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US7592409B2
- Patent Title: Styrene acrylonitrile isoprene triblock copolymer
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2009
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses controlled emulsion polymerization using a monomer solution containing a latent surfactant and a controlled free-radical polymerization agent, mixed with an aqueous medium containing a conjugate base to generate the emulsifier in situ. It is used to make block copolymers, including A-B-A architectures where the B block can contain acrylonitrile/methacrylonitrile and a conjugated diolefin such as 1,3-butadiene or isoprene. - Latent surfactant amount: ~0.2 grams — Example 1 - Styrene charge: 5 milliliters — Example 1 - Aqueous phase volume: 5 milliliters — Example 1 - Styrene charge: 1,000 grams — Example 2 - Oleic acid charge: 60.0 grams — Example 2 - Dibenzyltrithiocarbonate charge: 7.2 grams — Example 2 - RO water charge: 4,000 grams — Example 2 - Potassium persulfate charge: 40.0 grams — Example 2 - Tripotassium phosphate charge: 40.0 grams — Example 2 - Potassium hydroxide charge: 16.4 grams — Example 2 - Reaction temperature: 65 C — Example 2 - Complete conversion time: less than 1.5 hours — Example 2 - Styrene charge: 1,000 grams — Example 3 - Hexadecyl half ester of maleic acid charge: 60.0 grams — Example 3 - Dibenzyltrithiocarbonate charge: 7.2 grams — Example 3 - RO water charge: 4,000 grams — Example 3 - Potassium persulfate charge: 40.0 grams — Example 3 - Tripotassium phosphate charge: 40.0 grams — Example 3 - Potassium hydroxide charge: 16.4 grams — Example 3 - Reaction temperature: 65 C — Example 3 - Complete conversion time: less than 3 hours — Example 3 - Butadiene content in B block: 80.4 — Example 3 table - Acrylonitrile content in B block: 39.6 — Example 3 table - Butadiene content in B block: 71.3 — Example 3 table - Acrylonitrile content in B block: 35.2 — Example 3 table - Butadiene content in B block: 72.1 — Example 3 table - Acrylonitrile content in B block: 35.5 — Example 3 table - Butadiene content in B block: 60.9 — Example 3 table - Acrylonitrile content in B block: 29.8 — Example 3 table - Reaction temperature: 149 F — Example 3 table - RO water charge: 200 — Example 4 - RO water charge: 241 — Example 5 - RO water charge: 197.1 — Example 5 - RO water charge: 84.7 — Example 5 - RO water charge: 160.5 — Example 5 - Styrene charge: 50 grams (0.48 moles) — Example 8 - Oleic acid charge: 3.0 grams (0.0106 moles) — Example 8 - Phenyl-t-butyltrithione charge: 0.44 grams (0.002485 moles) — Example 8 - Distilled water charge: 200 grams — Example 8 - Potassium persulfate charge: 2.06 grams (0.00762 moles) — Example 8 - Tripotassium phosphate charge: 2.13 grams (0.01 moles) — Example 8 - Potassium hydroxide charge: 0.9 grams (0.014 moles) — Example 8 - Reaction temperature: 65 C — Example 8 -

Complete conversion time: about 7 hours — Example 8 - Styrene charge: 200 grams (1.923 moles) — Example 10 - Oleic acid charge: 12.0 grams (0.0424 moles) — Example 10 - Phenyl-t-butyltrinitrone charge: 2.0 grams (0.0113 moles) — Example 10 - Distilled water charge: 500 grams — Example 10 - Potassium persulfate charge: 8.0 grams (0.0296 moles) — Example 10 - Tripotassium phosphate charge: 8.0 grams (0.0377 moles) — Example 10 - Potassium hydroxide charge: 3.6 grams (0.056 moles) — Example 10 - Reaction temperature: 75 C — Example 10 - Complete conversion time: about 3.5 hours — Example 10 - Styrene charge: 50.0 grams (0.48 moles) — Example 13 - Oleic acid charge: 3.0 grams (0.0106 moles) — Example 13 - Benzaldehyde charge: 0.31 grams (0.00292 moles) — Example 13 - N-isopropylhydroxylamine solution: 1.4 grams (0.0028 moles) of 15% aqueous solution — Example 13 - Stirring time before styrene addition: 1 hour — Example 13 - Distilled water charge: 200 grams — Example 13 - Potassium persulfate charge: 2.06 grams (0.00762 moles) — Example 13 - Tripotassium phosphate charge: 2.13 grams (0.01 moles) — Example 13 - Potassium hydroxide charge: 0.9 grams (0.014 moles) — Example 13 - Reaction temperature: 75 C — Example 13 - Complete conversion time: about 4 hours — Example 13 - Styrene charge: 50.0 grams (0.48 moles) — Example 14 - Dodecylbenzenesulfonic acid charge: 3.0 grams (0.0092 moles) — Example 14 - Water charge: 1 milliliter — Example 14 - Sodium nitrite charge: 0.171 grams (0.002485 moles) — Example 14 - Distilled water charge: 200 grams — Example 14 - Potassium persulfate charge: 2.06 grams (0.00762 moles) — Example 14 - Tripotassium phosphate charge: 2.13 grams (0.01 moles) — Example 14 - Potassium hydroxide charge: 0.74 grams (0.0115 moles) — Example 14 - Reaction temperature: 75 C — Example 14 - Complete conversion time: about 3 hours — Example 14 - Styrene charge: 50 grams (0.48 moles) — Example 15 - Oleic acid charge: 3.0 grams (0.0106 moles) — Example 15 - 1,1-Diphenylethylene charge: 0.448 grams (0.002485 moles) — Example 15 - Distilled water charge: 200 grams — Example 15 - Potassium persulfate charge: 2.06 grams (0.00762 moles) — Example 15 - Tripotassium phosphate charge: 2.13 grams (0.01 moles) — Example 15 - Potassium hydroxide charge: 0.92 grams (0.0143 moles) — Example 15 - Reaction temperature: 75 C — Example 15 - Complete conversion time: about 2 hours — Example 15 - Styrene charge: 2,240 grams — Example 16 - Oleic acid charge: 134.4 grams — Example 16 - Dibenzyltrithiocarbonate charge: 16.0 grams — Example 16 - RO water charge: 3,808 grams — Example 16 - Potassium persulfate charge: 13.44 grams — Example 16 - Tripotassium phosphate charge: 89.6 grams — Example 16 - Potassium hydroxide charge: 36.96 grams — Example 16 - Reaction temperature: 65 C — Example 16 - Complete conversion time: less than 3 hours — Example 16 - Solids: 38.9% — Example 16 - Styrene charge: 50 grams (0.48 moles) — Example 17 - Sodium nitrite charge: 0.69 grams (0.01 moles) — Example 17 - Oleic acid charge: 3.0 grams (0.0106 moles) — Example 17 - Concentrated acetic acid charge: 1.0 grams (0.0167 moles) — Example 17 - Water charge: 2 grams — Example 17 - Distilled water charge: 160 grams — Example 17 - Potassium persulfate charge: 2.0 grams (0.00739 moles) — Example 17 - Tripotassium phosphate charge: 2.0 grams (0.01 moles) — Example 17 - Potassium hydroxide charge: 2.35 grams (0.0367 moles) — Example 17 - Reaction temperature: 75 C — Example 17 - Complete conversion time: about 2.5 hours — Example 17 - Solids: 26.1% — Example 17 - Styrene charge: 50 grams (0.48 moles) — Example 18 - Oleic acid charge: 3.0 grams (0.0106 moles) — Example 18 - 1-Chloro-1,1-diphenylethane: approximately 0.01 moles — Example 18 - Distilled water charge: 200 grams — Example 18 - Potassium persulfate charge: 2.0 grams (0.0074 moles) — Example 18 - Tripotassium phosphate charge: 2.0 grams (0.0094 moles)

— Example 18 - Potassium hydroxide charge: 1.55 grams (0.0242 moles) — Example 18 - Reaction temperature: 75 C — Example 18 - Complete conversion time: about 4 hours — Example 18 - Solids: 22.7% — Example 18 - Styrene charge: 1,300 grams (12.5 moles) — Example 19 - Oleic acid charge: 78.0 grams (0.2756 moles) — Example 19 - Crude 1-chloro-1,1-diphenylethane charge: 39.0 grams — Example 19 - Deionized RO water charge: 2,470 grams — Example 19 - Potassium persulfate charge: 50.7 grams (0.1875 moles) — Example 19 - Tripotassium phosphate charge: 50.7 grams (0.2388 moles) — Example 19 - Potassium hydroxide charge: 29.5 grams (0.602 moles) — Example 19 - Reaction temperature: 75 C — Example 19 - Complete conversion time: about 5.5 hours — Example 19 - Solids: 38.6% — Example 19 - 1-Hexene charge: 25 grams — Example 20 - n-Butyl acrylate charge: 25 grams — Example 20 - Dipyrityl charge: 0.38 grams — Example 20 - Oleic acid charge: 4.0 grams — Example 20 - Ethyl 2-bromoisobutyrate charge: 0.47 grams — Example 20 - 85% KOH charge: 1.7 grams — Example 20 - Water charge: 139 milliliters — Example 20 - Copper sulfate pentahydrate solution: 60.9 grams of 1% aqueous solution — Example 20 - Hydrazine hydrate: 2 drops — Example 20 - Reaction temperature: about 67 C — Example 20 - Reaction time: 1.5 hours — Example 20 - Solids: 10.9% — Example 20 - 1-Hexene incorporation: 5.5 mole% — Example 20

Relevant Experimental Evidence - Example 1: Demonstrated in-situ emulsification using latent surfactant/activator pairs to form emulsions before polymerization. - Example 2: Showed RAFT emulsion polymerization at 65 C with complete conversion in less than 1.5 hours and a stable latex. - Example 3: Demonstrated in-situ generated polymerizable surfactant and rapid microemulsion polymerization at 65 C. - Example 5: Showed preparation of S-NBR-S triblock copolymer latexes from seed latexes. - Example 20: Demonstrated microemulsion ATRP copolymerization and confirmed 5.5 mole% 1-hexene incorporation.

Technical Relevance This patent is relevant because it discloses controlled emulsion polymerization methods that can accommodate nitrile-containing blocks and block architectures, which are useful for designing NBR-based materials with tailored acrylonitrile distribution. Although it is not a direct low-ACN NBR synthesis patent, its in-situ emulsification and seed/block strategies are transferable to controlled NBR latex synthesis.

Patent 2

- Patent Number: US5770660A
- Patent Title: Continuous polymerization process for producing NBR rubber having a high bound ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 1998
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses a continuous free-radical emulsion polymerization process for acrylonitrile and butadiene in which butadiene is added incrementally across multiple reaction vessels. The process is designed to maintain a substantially constant unconverted monomer ratio and produce NBR with high bound acrylonitrile and improved tensile/environmental resistance properties. - Initial monomer ratio: about 80 parts acrylonitrile to 20 parts $\pm 10\%$ butadiene — Claim 1 - Unconverted monomer ratio: about 70 to 90 parts acrylonitrile to about 30 to 10 parts butadiene — Claim 1 - Bound acrylonitrile content: about fifty percent by weight — Claim 1 - Example I initial butadiene charge: 21 parts — Example I - Example I acrylonitrile charge: 54 parts — Example I - Example I conversion: 50.2 percent — Example I - Example I bound acrylonitrile: 59.5 percent by weight — Example I - Example II initial butadiene charge: 21 parts — Example II - Example II acrylonitrile charge: 54 parts — Example II - Example II additional butadiene charge: 16 parts — Example II - Example II conversion: 73.3 percent — Example II - Example II bound acrylonitrile: 52.6 percent by weight — Example II - Example III conversion: 73.6 percent — Example III - Example III bound acrylonitrile: 54.9 percent — Example III - Example IV initial butadiene charge: 21 parts — Example IV - Example IV additional butadiene charge: 16 parts — Example IV - Example IV additional butadiene charge: 9 parts — Example IV - Example IV conversion: 85.2 percent — Example IV - Example IV bound acrylonitrile: 50.5 percent — Example IV - Example V acrylonitrile charge: 56 parts — Example V - Example V total butadiene charge: 43 parts — Example V - Example V initial butadiene charge: 22 parts — Example V - Example V additional butadiene charges: 8 parts, 7 parts, 6 parts — Example V - Example V final conversion: 81.9 percent — Example V - Example V bound acrylonitrile: 50.2 percent — Example V - Example VI final conversion: 86.8 percent — Example VI - Example VI bound acrylonitrile: 50.6 percent — Example VI - Example VII acrylonitrile charge: 56 parts — Example VII - Example VII initial butadiene charge: 20 parts — Example VII - Example VII additional butadiene charges: 6 parts, 6 parts, 6 parts, 6 parts — Example VII - Example VII steady-state conversion in first vessel: 24 percent — Example VII - Example VII steady-state conversion in second vessel: 40 percent — Example VII - Example VII steady-state conversion in third vessel: 56 percent — Example VII - Example VII steady-state conversion in fourth vessel: 72 percent — Example VII - Example VII steady-state conversion in fifth vessel: 82 percent — Example VII - Example VII bound acrylonitrile: 48.8 percent — Example VII

Relevant Experimental Evidence - Example I: Conventional batch process with limited butadiene gave 59.5 wt% bound acrylonitrile but only 50.2% conversion. - Example II: Two-increment butadiene addition improved conversion to 73.3% and produced 52.6 wt% bound acrylonitrile. - Example IV: Three butadiene increments raised conversion to 85.2% while maintaining 50.5% bound acrylonitrile. - Example V: Four butadiene increments with 56 parts acrylonitrile and 43 parts butadiene gave 50.2 wt% bound acrylonitrile at 81.9% conversion. - Example VII: Continuous five-vessel operation maintained staged conversions and produced 48.8% bound acrylonitrile with improved tensile and environmental resistance.

Technical Relevance This patent is highly relevant because it directly addresses acrylonitrile/butadiene copolymerization and shows how staged butadiene feed controls composition drift, a key issue in low-ACN and composition-tuned NBR. The continuous feed strategy and monomer-ratio management provide practical guidance for producing more uniform

NBR latexes with targeted acrylonitrile content.

Patent 3

- Patent Number: US3426102A
- Patent Title: Polymerizates of olefinic nitriles and diene-nitrile rubbers
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 1969
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses preparation of a preformed nitrile rubber latex by emulsion polymerization of acrylonitrile and butadiene under agitation, low oxygen, and controlled pH, followed by use of that latex as a rubber phase in impact-resistant nitrile polymer compositions. The examples also show grafting/latex blending with acrylonitrile-containing monomer systems to form high-impact barrier resins. - Example I-A acrylonitrile charge: 60 parts — Example I - Example I-A butadiene-1,3 charge: 40 parts — Example I - Example I-A emulsifier charge: 2.4 parts GAFAC RE-610 — Example I - Example I-A initiator charge: 0.3 parts azo-bis-isobutyronitrile — Example I - Example I-A chain-transfer agent charge: 0.5 parts t-dodecyl mercaptan — Example I - Example I-A water charge: 200 parts — Example I - Example I-A polymerization temperature: 45 C — Example I - Example I-A pH: about 8 — Example I - Example I-A reaction time: 22.5 hours — Example I - Example I-A conversion: 92% — Example I - Example I-A total solids: 33.1% — Example I - Example I-B acrylonitrile charge: 70 parts — Example I - Example I-B ethyl acrylate charge: 30.0 parts — Example I - Example I-B latex A charge: 31.9 parts — Example I - Example I-B initiator charge: 0.06 parts potassium persulfate — Example I - Example I-B emulsifier charge: 3.0 parts GAFAC RE-610 — Example I - Example I-B chain-transfer agent charge: 1.0 parts n-dodecyl mercaptan — Example I - Example I-B chelating agent charge: 0.05 parts ethylene diamine tetracetic acid — Example I - Example I-B water charge: 200 parts — Example I - Example I-B pH: about 7 — Example I - Example I-B polymerization temperature: 60 C — Example I - Example I-B reaction time: 20 hours — Example I - Example I-B conversion: 97% — Example I - Example I-B total solids: 33% — Example I - Example I-C acrylonitrile charge: 73.6 parts — Example I - Example I-C ethyl acrylate charge: 30 parts — Example I - Example I-C butadiene charge: 5.4 parts — Example I - Example I-C total solids: 38.3% — Example I - Example I-C conversion: 87.7% — Example I - Example 11-A acrylonitrile charge: 40 parts — Example 11 - Example 11-A butadiene charge: 60 parts — Example 11 - Example 11-A emulsifier charge: 2.4 parts — Example 11 - Example 11-A potassium persulfate charge: 0.1 parts — Example 11 - Example 11-A t-dodecyl mercaptan charge: 0.5 parts — Example 11 - Example 11-A water charge: 200.0 parts — Example 11 - Example 11-A polymerization temperature: 45 C — Example 11 - Example 11-A pH: about 8 — Example 11 - Example 11-A reaction time: 5.5 hours — Example 11 - Example 11-A conversion: 97% — Example 11 - Example 11-A total solids: 33% — Example 11 - Example 11-B acrylonitrile

charge: 70 parts — Example 11 - Example 11-B ethyl acrylate charge: 30 parts — Example 11 - Example 11-B latex A charge: 33 parts — Example 11 - Example 11-B potassium persulfate charge: 0.06 parts — Example 11 - Example 11-B n-dodecyl mercaptan charge: 1.0 parts — Example 11 - Example 11-B emulsifier charge: 3.0 parts — Example 11 - Example 11-B ethylene diamine tetraacetic acid charge: 0.05 parts — Example 11 - Example 11-B water charge: 200.0 parts — Example 11 - Example 11-B polymerization temperature: 60 C — Example 11 - Example 11-B reaction time: 20 hours — Example 11 - Example 11-B conversion: 96% — Example 11 - Example 11-B total solids: 33% — Example 11 - Example V acrylonitrile charge: 20 parts — Example V - Example V ethyl acrylate charge: 20 parts — Example V - Example V rubber latex charge: 50 parts — Example V - Example VI rubber latex charge: 20 parts per parts of acrylonitrile-ethyl acrylate monomer mixture — Example VI - Example VII rubber content: about 9 parts by weight — Example VII - Example VIII rubber content: 12 parts per 112 parts of polymer — Example VIII - Example XI-A methacrylonitrile charge: sole monomer — Example XI - Example XI-A t-butyl peroxyphthalate charge: 0.4 part — Example XI - Example XI-A n-dodecyl mercaptan charge: 0.5 part — Example XI - Example XI-A emulsifier charge: 2.4 parts — Example XI - Example XI-A water charge: 225 parts — Example XI - Example XI-A polymerization temperature: not explicitly stated in the example text beyond the procedure reference — Example XI - Example XI-A rubber latex loading: 10 parts of 25:75 acrylonitrile:butadiene rubber per 110 parts total polymer — Example XI - Example XII-A methacrylonitrile charge: 90 parts — Example XII - Example XII-A ethyl acrylate charge: 10 parts — Example XII - Example XII-A t-butyl peroxyphthalate charge: 0.3 part — Example XII - Example XII-A n-dodecyl mercaptan charge: 0.5 part — Example XII - Example XII-A emulsifier charge: 2.4 parts — Example XII - Example XII-A water charge: 200 parts — Example XII - Example XII-A polymerization temperature: 60 C — Example XII - Example XII-A pH: about 7 — Example XII

Relevant Experimental Evidence - Example I-A: Prepared a 60/40 acrylonitrile/butadiene rubber latex at 45 C with 92% conversion and 33.1% solids. - Example I-B: Grafted 70/30 acrylonitrile/ethyl acrylate onto the NBR latex at 60 C to form a high-impact barrier resin. - Example I-C: Showed a terpolymer containing acrylonitrile, ethyl acrylate, and butadiene with 87.7% conversion. - Example 11-A: Repeated the NBR latex preparation at 45 C using 40/60 acrylonitrile/butadiene and obtained 97% conversion. - Example XI-A: Demonstrated use of a 25:75 acrylonitrile/butadiene rubber latex in methacrylonitrile-based graft polymerization.

Technical Relevance This patent is relevant because it explicitly discloses acrylonitrile/butadiene rubber latexes with defined monomer ratios, including a 40/60 acrylonitrile/butadiene example that falls within the low-to-moderate ACN NBR range. It also shows how such latexes can be used as preformed rubber phases in aqueous polymerization, which is important for understanding low-ACN NBR preparation and utilization.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- US5770660A is the only patent in the set that directly targets acrylonitrile/butadiene copolymerization as the primary product; it uses continuous staged butadiene addition to control monomer ratio drift and achieve about 48.8–59.5 wt% bound acrylonitrile.
- US3426102A discloses preformed NBR latexes with explicit acrylonitrile/butadiene ratios of 60/40 and 40/60, plus a 25/75 rubber latex used in graft/impact resin systems; its process is batch emulsion polymerization at 45–60 C with emulsifier, initiator, and mercaptan chain transfer agent.
- US7592409B2 is mechanistically relevant but not a direct NBR patent; it teaches in-situ emulsifier formation and controlled emulsion/block polymerization, including B blocks containing acrylonitrile and butadiene/isoprene, which could be adapted to controlled NBR architectures.
- Compared with the older batch emulsion route in US3426102A, US5770660A provides stronger composition control by staged monomer feeding and multiple reactors, reducing the monomer-ratio swing that normally occurs as butadiene is depleted.
- US3426102A uses conventional emulsion ingredients such as GAFAC RE-610, potassium persulfate or azo-bis-isobutyronitrile, and t-dodecyl/n-dodecyl mercaptan; US5770660A emphasizes process staging rather than a novel catalyst system.
- US7592409B2 operates at 65–75 C in microemulsion/RAFT/NMP-like systems with oleic acid, potassium persulfate, tripotassium phosphate, and potassium hydroxide, whereas US5770660A and US3426102A are conventional NBR emulsion processes centered on monomer feed and latex stability.
- For low-acrylonitrile NBR specifically, the most transferable lesson is from US5770660A: control the acrylonitrile/butadiene feed ratio and avoid complete butadiene depletion to preserve uniform composition.
- US3426102A suggests that low-to-moderate ACN rubber phases can be prepared at 45 C with about 8 pH and 33% solids, which may be useful for low-ACN latex design and downstream blending.

4. CONCLUSION

The patent evidence shows that low-acrylonitrile NBR synthesis is best approached through emulsion polymerization with tight monomer-feed control and stable latex formation. The strongest direct guidance comes from US5770660A, which demonstrates that staged butadiene addition across multiple reactors can maintain a more uniform acrylonitrile/butadiene ratio and avoid the composition drift that degrades properties in conventional batch NBR. US3426102A provides older but still useful evidence that acrylonitrile/butadiene latexes can be made at 45 C with emulsifier, initiator, and mercaptan chain-transfer control, including a 40/60 acrylonitrile/butadiene rubber latex that is within the low-to-moderate ACN regime. US7592409B2 is not a direct NBR patent, but it supports controlled emulsion strategies and block/seed architectures that could be adapted to low-ACN nitrile rubber systems. Based on the evidence, recommended synthesis parameters are emulsion polymerization, staged or incremental monomer feed, moderate temperatures around 45–60 C for conventional NBR latex routes, and careful maintenance of unconverted monomer

ratio to prevent butadiene depletion and composition drift.

5. REFERENCES

- US7592409B2 | Styrene acrylonitrile isoprene triblock copolymer | Not disclosed | US | 2009 | <https://patents.google.com/patent/US7592409B2/en>
- US5770660A | Continuous polymerization process for producing NBR rubber having a high bound ... | Not disclosed | US | 1998 | <https://patents.google.com/patent/US5770660A/en>
- US3426102A | Polymerizates of olefinic nitriles and diene-nitrile rubbers | Not disclosed | US | 1969 | <https://patents.google.com/patent/US3426102A/en>